DATA MATRIX ANALYSIS

AP

MOTIVATIONS

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Activity tables show how users *map* their choices,

or how available products *map* onto their adopters.

	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

Figure 11.6: Ratings of movies by users

	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
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John	4	4	4	0	0
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Figure 11.6: Ratings of movies by users

Data Science view: 7 user profiles; 18 datapoints of user review.

Network science view: a weighted, binary relationship between users and films.

Let's step back and see what Geometry gives us.

SPECTRAL METHODS

Most activity matrices represent the connections between n entities (e.g., subscribers) and m entities (e.g., films) with $n <> m$.

Sometimes the connections is between the same entities, such as endorsement, teams defeating other teams, friends or followers on social networks etc.

In such cases, the matrix is *square*.

Standard Geometry holds and we can extract its *eigenpairs*.

MATRICES

TRADITIONAL VIEW

There is a *space* in *n* dimensions

Each point is represented by an n-dimensional vector $\mathbf{x}$ (or $\vec{x}$)

Transformation: combination of *expansion* (or a *contraction*,) and *rotation*, is represented by a matrix, A.

The resulting point is computed as $A\mathbf{x}$.

EXAMPLE

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

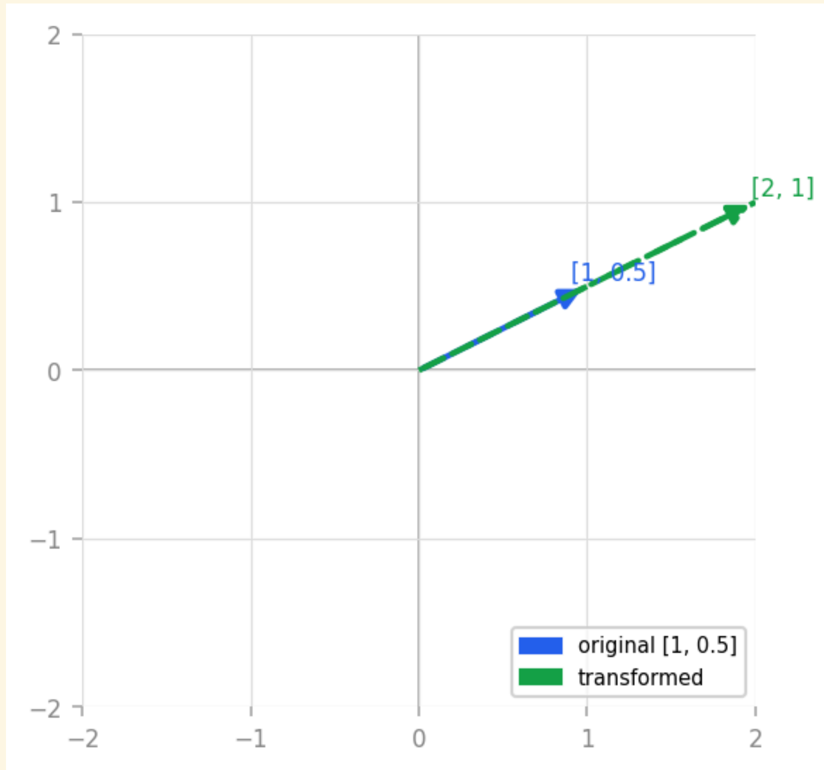
$$M\mathbf{x} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

M does anti-clockwise rotation and expansion at once.

MORE EXAMPLES

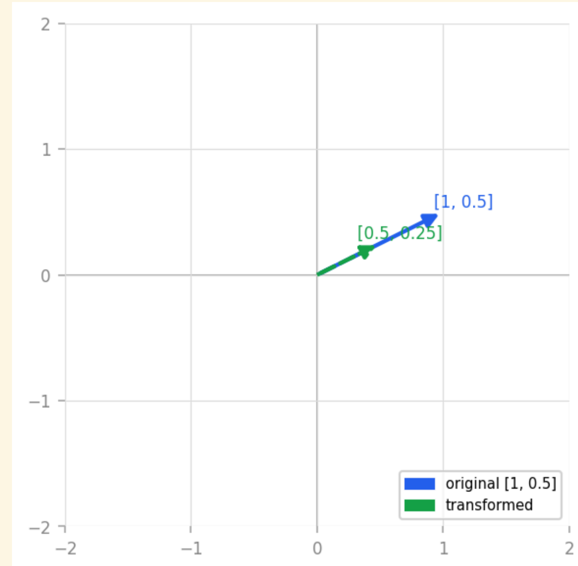
expand $\mathbf{x}$ to double its length:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$



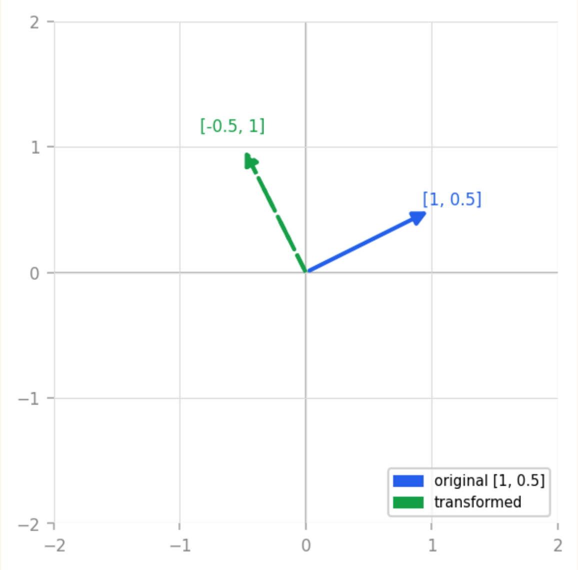
compress $\mathbf{x}$ to halve its length:

$$\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.25 \end{pmatrix}$$



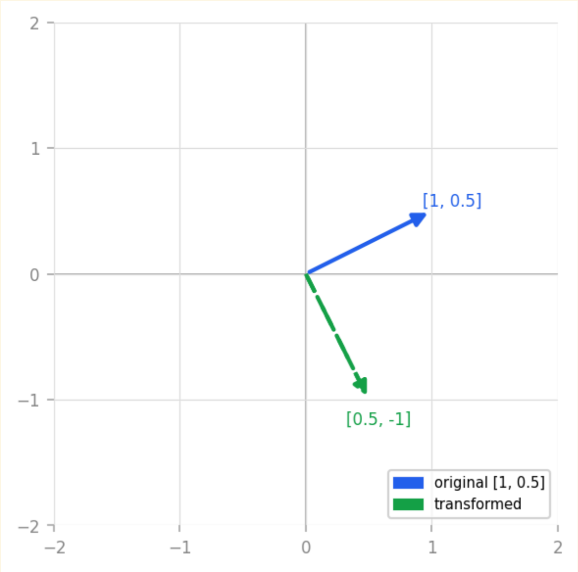
rotate $\mathbf{x}$ anticlockwise:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 1 \end{pmatrix}$$



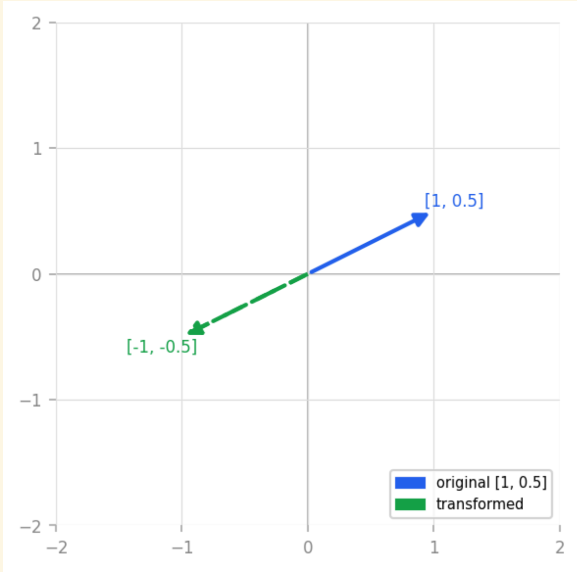
rotate $\mathbf{x}$ clockwise:

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -1 \end{pmatrix}$$



change sense/flip:

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} -1 \\ -0.5 \end{pmatrix}$$



EIGENPAIRS

DEFINITION

If matrix A has a real λ and a vector $\mathbf{e}$ s.t.

$$A\mathbf{e} = \lambda\mathbf{e}$$

then λ is an *eigenvalue* and $\mathbf{e}$ an *eigenvector* of A.

If *A* has rank *n* then there could be up to *n* eigenpairs.

- However, they might not be real, nor $\neq 0$, and
- extracting them is *costly*: at least n^2 scalar multiplications, n being the size of the matrix.

Example: the traditional 90-degrees rotation matrix:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has *unplayable* eigenpairs:

$$\lambda_1 = i, \mathbf{e_1} = [1, -i]^T$$

$$\lambda_2 = -i, \mathbf{e_2} = [1, i]^T$$

EIGENDECOMPOSITION

The *effect* of A can be decomposed into basic expansion/contraction and rotation thanks to the extraction of *eigenpairs* $\langle \boldsymbol{\lambda}, \mathbf{e} \rangle$:

$$A\mathbf{e} = \boldsymbol{\lambda}\mathbf{e}$$

Consider the M matrix again:

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

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solve $M\mathbf{e} = \boldsymbol{\lambda}\mathbf{e}$ to find

$\lambda_1 \approx 5.372$ and $\lambda_2 \approx -0.372$.

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$$\mathbf{e_1} \approx \begin{pmatrix} 0.416 \\ 0.909 \end{pmatrix}$$

$$\mathbf{e_2} \approx \begin{pmatrix} 0.824 \\ -0.0566 \end{pmatrix}$$

Since its eigenpairs are real, M can be *diagonalised:* re-formulated as the product of simpler `operations'

$$M = Q\Lambda Q^{-1}$$

Rotation (Q^{-1}) followed by expansion (Λ) followed by 'rotation back' (Q)

$$Q = \begin{pmatrix} \mathbf{e_1} & \mathbf{e_2} \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

FINALLY

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} =$$

$$\begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.575 & 0.837 \\ 0.924 & -0.423 \end{pmatrix}$$

EXAMPLE APPLICATION

$$M\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

now becomes

$$\begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.575 & 0.837 \\ 0.924 & -0.423 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

$$
\begin{aligned}
M\mathbf{x} &= \begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.994 \\ 0.712 \end{pmatrix} \\
&= \begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.340 \\ -0.265 \end{pmatrix} \\
&= \begin{pmatrix} 2.003 \\ 5.004 \end{pmatrix}
\end{aligned}
$$

numerical issues start arising

NOTABLE SQUARE M.

A is called *symmetric* when $A = A^T$

Our M is *not* symmetric:

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = M^T$$

Theorem: symmetric m. have all eigenpairs real: they can always be diagonalised

A is called *positive semidefinite* when, for any $\mathbf{x}$, we have

$$\mathbf{x}^T A\mathbf{x} \geq 0$$

In such case its eigenvalues are guaranteed non-negative: $\lambda_i \geq 0$, this helps interpretation of activity matrices.

SINGULAR-VALUE DECOMPOSITION

Eigendecomposition is defined only for square matrices.

SVD will provide a similar decomposition for data matrices of any size.

Details are in the [Goodfellow et al.] textbook and will be introduced later on.

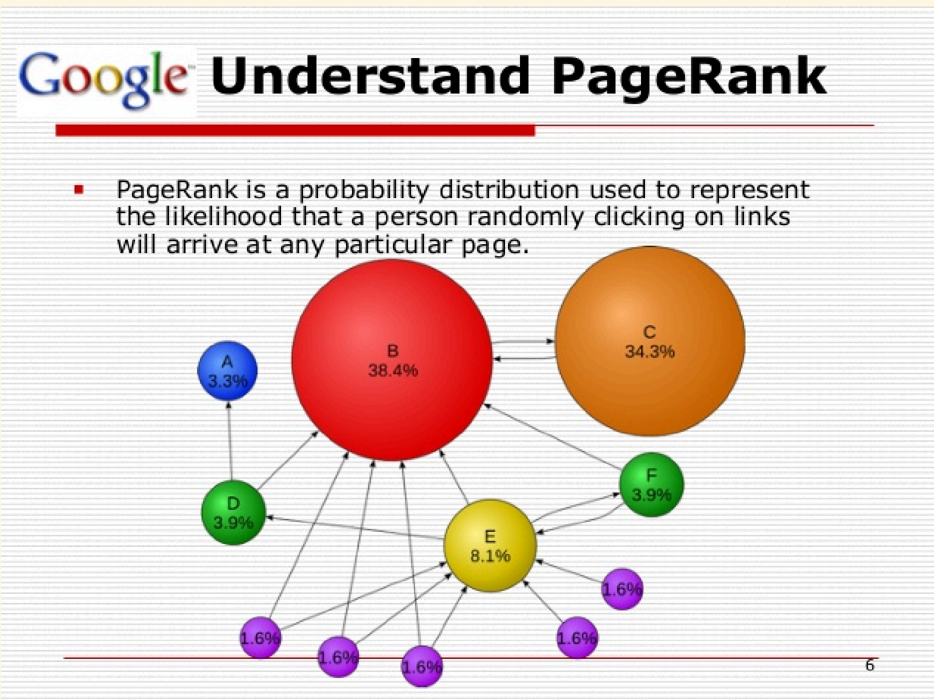
APPLICATIONS

SPECTRAL PROPERTIES

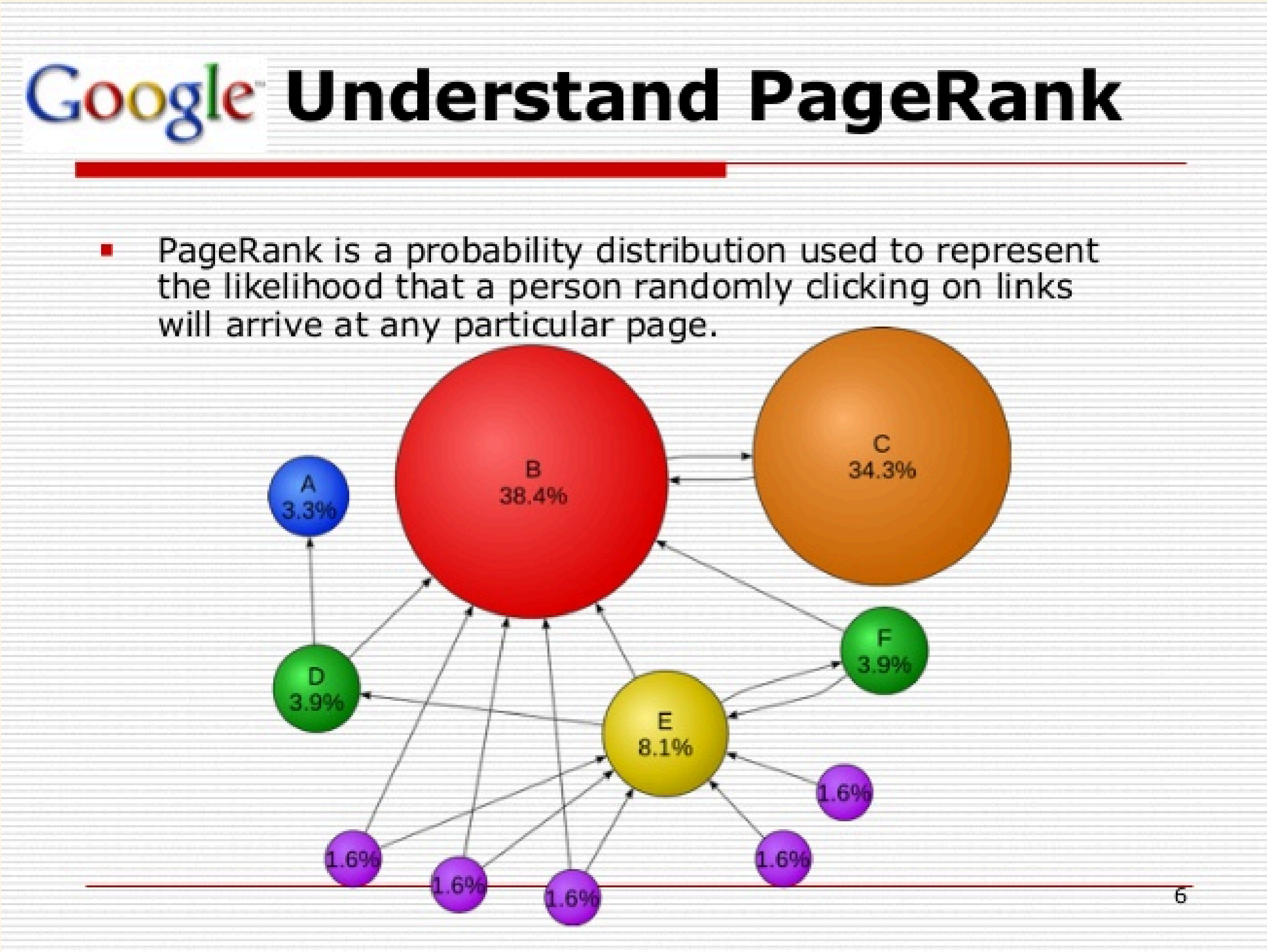
Adjacency matrices represent connections between entities in a network (graph), e,g., the Web.

The eigenvalues of adjacency matrices provide bounds for several network features.

Google PageRank algorithm *is* spectral network anal.



Web page importance ≈ eigenvector centrality:



Early applications in Psychology, Social science, Bibliometrics, Economics, and Choice theory (seriously).

SPECTRAL RANKING

Given a matrix representing preference or likeability between people, can we rank the participants (from best to worst) on the basis of their general, intrinsic likeability?

[Seely, 1949] created an index of likeability based on the ideas of *diffusion*: it is important to be liked by people who in turn are well-liked and so on.

Let M be a square matrix where m_{ij} represents *approval* or *endorsement* (negative values represent *disapproval)*

my *likeability index* should be equal to the weighted sum of of the indices of the people who like me.

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But their likeability is turn will depend on mine...

Let's use row vectors $\mathbf{r} = [r_1, r_2, \ldots r_n]$:

$$\mathbf{r} = \mathbf{r}M$$

i.e., $\mathbf{r}$ is a left eigenvector of M.

This formula might have no solution, but matrix preprocessing can assure that one exists.

STUDY PLAN

BACKGROUND STUDY

Ian Goodfellow, Yoshua Bengio and Aaron Courville: Deep Learning, MIT Press, 2016.

available in HTML and PDF; it is *a refresher* of notation and properties: no examples and no exercises.

It can be read in the background.

- Phase 1: read §§ 2.1—2.7, then § 2.11.
- Phase 2: read §§ 2.8—2.10